Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of IEC Standard-based Add-on AI Edge Monitoring and Security Module for Heterogeneous Power Converters**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Heavy Electrical Equipment	*Transmission/Distribution, Protection and Monitoring Devices*		
Concept of Proposed Technology	**Core Concept** This technology aims to develop a PCB-type hardware module that can be added on to existing power conversion devices without altering their internal circuits. This module integrates sensor circuits, an Edge AI processor, and a security chip (TPM) on a single board to perform the following functions. **Key Functions and Potential Applications** ● Multi-physics Sensing: Collects data via non-intrusive current/voltage sensors attached to power lines, vibration sensors on the equipment enclosure, and EMI sensors detecting internal arcs and noise. ● AI-based Integrated Diagnosis: Analyzes collected data at the edge to perform performance			

	monitoring, anomaly detection, and fault diagnosis. It specifically diagnoses efficiency drops in PV inverters, insulation degradation in Solid State Transformers (SST), and component stress due to transient loads in AI server Power Supply Units (PSU). ● Standard Security Gateway: Implements IEC 62351 (Communication Security) and IEC 62443 (System Security) standards to encrypt communication data and ensure integrity for legacy equipment lacking built-in security.
Rationale for Joint R&D Project	**Critical Technical Barriers** ● Non-intrusive Sensing Without Modifying Internal Circuits ■ Existing PV inverters, SSTs, and AI server PSUs have sealed, vendor-specific internal circuitry, making it extremely difficult to diagnose arc faults, insulation degradation, or transient stress without opening the device. Achieving high-fidelity measurements using only externally attached non-intrusive sensors remains a major technical challenge. ● Edge AI Diagnosis Under High-Noise, High-Frequency Switching Environments ■ Power converters operate in harsh electromagnetic environments with MHz-level switching noise from SiC/GaN devices. Maintaining over 95% F1-score using lightweight TinyML models on resource-constrained edge processors is technically demanding due to severe signal distortion and noise contamination. ● Real-time Monitoring Requirements ■ Critical failure modes such as arc faults and insulation breakdown evolve within tens of milliseconds. At the same time, implementing IEC 62351/62443-compliant encryption must keep communication latency below 10 ms. Balancing real-time anomaly detection with secure communication on low-power hardware is a significant technical barrier. ● Integration of IEC-standard Security into Legacy Power Equipment ■ Legacy PV/SST/PSU equipment generally lacks cybersecurity functionality or uses proprietary, non-standard communication protocols. Converting these protocols (Modbus, DNP3, etc.) to IEC 61850/62351 formats while ensuring data consistency and device interoperability requires advanced firmware engineering and deep knowledge of power-system communication standards. **Why Global Collaboration?** International collaboration is essential to combine source technology acquisition with commercialization capabilities. ● Purdue University (USA): Possesses world-class power electronics research facilities (CPERC, Grainger Lab), providing core source technologies for failure mechanism analysis of PV/SST, Edge AI algorithm (F/W) design, and hardware circuit (Schematic) design. ● Korean SME: Based on Purdue's designs, the Korean partner is responsible for industrial PCB manufacturing suitable for field application and utilizes extensive testbeds (PV plants, Data Centers) for productization and reliability verification (Achieving TRL 7).

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Key Words	1. Add-on Edge Module 2. Prognostics and Health Management (PHM) 3. IEC 62443/62351 (Industrial Security Standards) 4. Solid State Transformer (SST) 5. AI Server PSU

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

This technology aims to develop a PCB-type hardware module that can be added on to existing power conversion devices without altering their internal circuits. This module integrates sensor circuits, an Edge AI processor, and a security chip (TPM) on a single board.

Detailed Objectives

- Develop a standardized Add-on PCB hardware platform compatible with PV inverters, SSTs, and AI server PSUs.
- Develop lightweight AI diagnosis firmware (F/W) to diagnose device-specific failure characteristics (Arc, Insulation breakdown, Load transients).
- Implement secure communication firmware (F/W) that converts legacy protocols (Modbus, DNP3) to IEC 61850 and applies encryption.
- Advance Technology Readiness Level (TRL) from Level 3 (Proof of Concept) at the start to Level 7 (System Prototype Demonstration in Operational Environment) by completion.

R&D Target Specifications

Category	Item	Target Specification (Quantitative)	Note
Monitoring	Diagnosis Accuracy	Over 95% (F1-Score)	AI Model
Monitoring	Anomaly Detection Speed	Within 100ms	Edge Inference
Security	Security Standard	IEC 62443 SL-2	Certification
Security	Encryption Latency	Less than 10ms	Real-time
H/W	Sampling Rate	Over 1 MHz (High-freq noise)	Sensor Circuit
H/W	Reliability	P65, -20~60°C Operation	Industrial Grade

R&D Plan

The research will be conducted over a total period of 3.5 years (July 2026 ~ June 2029), including a 6-month pre-research phase.

- Pre-research (2026.07.01 ~ 2026.12.31, 6 Months): Finalize Requirement Definition and Baseline Foundation (TRL 3)
  - [Common] Hold Korea-US joint technical workshops and finalize User Requirement Definition (URD).
  - [Purdue] Analyze latest trends in fault diagnosis and review non-intrusive sensor

	candidates (Hall, Piezo, RF). ■ [Korea SME] Analyze testbed environments (PV, Data Center), survey Grid Codes/regulations, and plan component supply. ● Year 1 (2027, 12 Months): Core Design & Hardware Fabrication (TRL 3→4) ■ [Purdue] Design Add-on module circuits (Schematic) and generate virtual fault data for AI training. ■ [Korea SME] Select Sensors/MCU, perform PCB Artwork, and manufacture 1st test boards. ● Year 2 (2028, 12 Months): Prototype Development & Lab Testing (TRL 5) ■ [Purdue] Implement lightweight Edge AI (TinyML) models and IEC 61850/62351 security F/W. ■ [Korea SME] Apply IP65 housing design, manufacture integrated prototypes, and conduct unit functional testing. ● Year 3 (Jan 2029 ~ June 2029, 6 Months): Field Demonstration & Commercialization (TRL 6→7) ■ [Korea SME] Install at PV plants and Data Centers, collect long-term operational data. ■ [Purdue] Fine-tune AI models based on field data and conduct security vulnerability assessments (Red Teaming).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Technical Contribution ■ Provide source technologies for sensing algorithms and AI model architecture based on Physics of Failure. ● Design Tasks ■ H/W Design: Design schematics for sensor interfaces and signal processing units, guide major chipset selection. ● F/W Development ■ Train and lightweight (TinyML) AI models for anomaly detection, implement IEC 62351/62443 security stacks. ● Verification ■ Conduct simulations and lab-level algorithm verification using facilities like the Grainger Lab.
Expected R&R of Korean SME Partner	● Development Tasks (Manufacturing) ■ PCB Layout and noise countermeasure design based on Purdue's schematics. ■ Mechanical Design (Housing) and mold production considering field conditions (Compact/Waterproof/Heat dissipation). ■ Manufacture final Add-on module prototypes and review mass production feasibility. ● Test/Verification Activities (Demonstration) ■ Provide Testbeds and install modules at domestic PV plants, ESS centers, and AI data centers. ■ Conduct durability tests through long-term operation and pursue national certifications (KC/CE). ■ Collect field data suitable for the Korean grid environment and provide feedback.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Technical Achievement ■ Secure core 'Retrofit' technology to convert legacy power equipment into intelligent systems without replacement. Preempt high-value module technology fusing Edge AI and international security standards. ● Economic Impact ■ Establish a continuous monitoring system at 1/10th the cost of expensive diagnostic equipment. Prevent massive economic losses by improving PV efficiency and preventing AI data center downtime. ● Commercial Potential ■ US Market Entry: Early entry into the North American market with products meeting strengthened US grid security regulations (IEC standards) through collaboration with Purdue. ■ New Business Creation: Expand business models beyond hardware sales to 'Subscription-based Predictive Maintenance Service (SaaS)' using analyzed data.

6. References

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